

Project Interim Report

On

Soil Health & Fertilizer Recommendation System

(Using Machine Learning, Flask & Weather API)

RU-500-02-00002

Foundations of Machine Learning

SESSION: 2025-26



Submitted By

ASHPREET KAUR

RU-25-10317

**Master of Computer Application in AI and ML in
Association with Google**

Under the Guidance of

Mr. Zamin Hamid

SCHOOL OF COMPUTER INFORMATION TECHNOLOGY

RUNGTA INTERNATIONAL SKILLS UNIVERSITY

BHILAI, CG

(April 2026)

Abstract

The Soil Health and Fertilizer Recommendation System is a Machine Learning and Flask-based web application designed to help farmers make smart agricultural decisions. The system predicts the most suitable fertilizer based on soil nutrient values such as nitrogen, phosphorus, potassium, soil pH, crop type, and crop growth stage.

The application also integrates real-time weather information using the OpenWeatherMap API to provide climate-based advisory recommendations. The project aims to reduce manual fertilizer selection errors and improve crop productivity through data-driven recommendations.

The system includes soil health analysis, fertilizer cost estimation, weather dashboard, and explainable AI reasoning for better understanding of recommendations. The frontend of the application is developed using HTML, CSS, Bootstrap, and Jinja templates, while Flask is used as the backend framework. The Random Forest Machine Learning algorithm is used for prediction because of its accuracy and efficiency.

Introduction

Agriculture plays an important role in the economy, and fertilizer management is one of the most critical factors for improving crop yield and soil health. Farmers often face challenges in selecting the correct fertilizer based on changing soil conditions and climate variations. The Soil Health and Fertilizer Recommendation System is developed to solve this problem using Machine Learning and real-time weather analysis.

The system takes input such as soil pH, nitrogen level, phosphorus level, potassium level, crop type, and crop growth stage. It then uses a trained Random Forest Machine Learning model to predict the best fertilizer. Additionally, the application provides soil health status, estimated fertilizer cost, weather insights, and climate advisory messages.

Objectives

- Predict suitable fertilizer using Machine Learning
- Analyze soil health conditions
- Provide real-time weather intelligence
- Reduce manual farming decisions
- Improve crop productivity
- Display explainable AI recommendations
- Provide estimated fertilizer cost

Scope

This project is suitable for smart agriculture applications and can help farmers make better fertilizer decisions. The system can be extended in future with IoT sensors, mobile applications, multilingual support, cloud deployment, and AI chatbot integration.

System Requirements

- **Hardware Requirements:**

Computer/Laptop with minimum 4GB RAM
Internet Connection

- **Software Requirements:**

Python 3.x
Flask Framework
VS Code / PyCharm
OpenWeatherMap API

Methodology

1. User Input Collection

The user enters soil details, crop information, and city name through the web interface.

2. Weather Data Collection

The system fetches real-time weather data such as temperature, humidity, and rainfall using the Weather API.

3. Data Preprocessing

The input data is cleaned, encoded, and scaled using StandardScaler before prediction.

4. Machine Learning Prediction

The Random Forest Machine Learning model predicts the most suitable fertilizer.

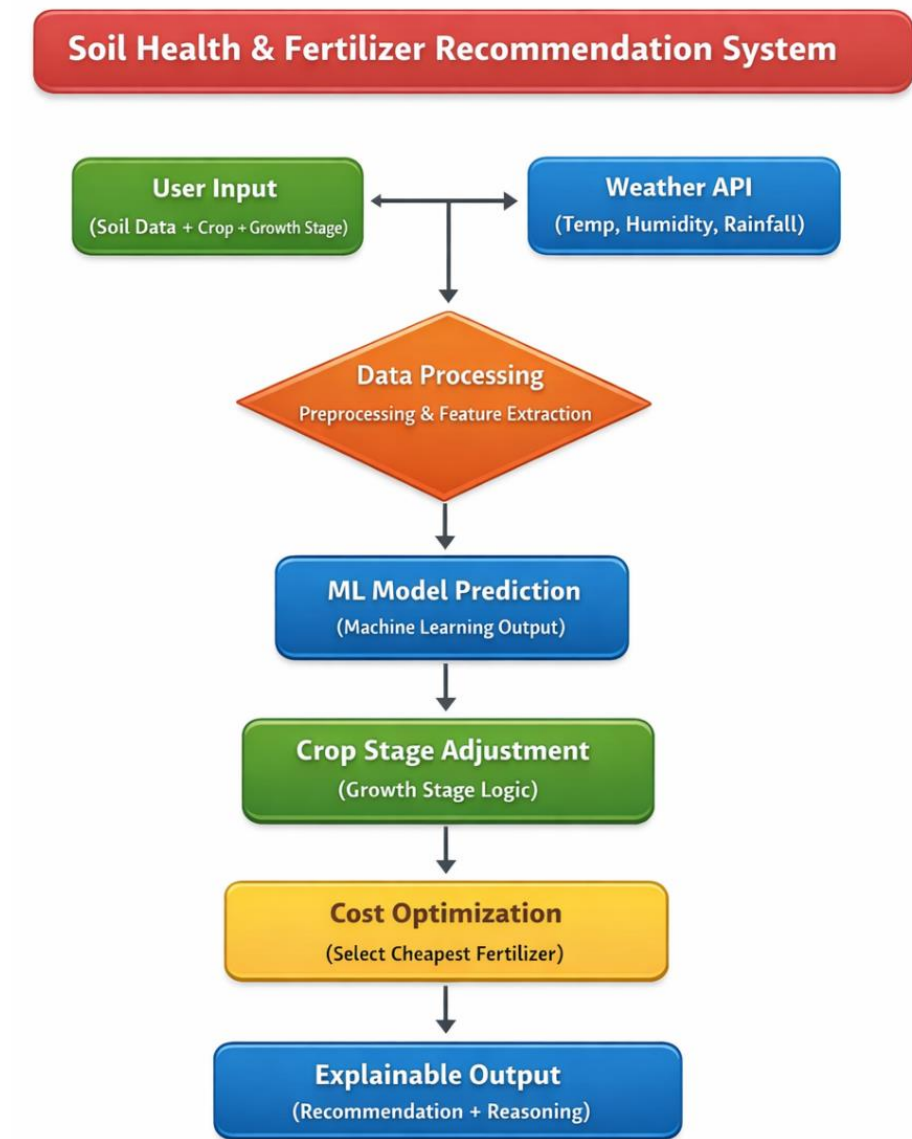
5. Soil Health & Climate Analysis

The system generates soil health status and climate advisory messages based on soil and weather conditions.

6. Result Display

The final fertilizer recommendation and analysis are displayed on the dashboard.

Flowchart



Implementation

The project consists of multiple Python modules. The preprocessing module handles dataset cleaning, encoding, and feature scaling. The model training module trains the Random Forest classifier and stores trained models using pickle. The Flask application handles routing, user input, weather API integration, prediction generation, and result rendering. The frontend is developed using Bootstrap and custom CSS for responsive dashboard design.

Sample Input/Output

Sample Input:

City: Raipur
Soil pH: 6.5
Nitrogen: 45
Phosphorus: 50
Potassium: 60
Crop Type: Rice
Growth Stage: Vegetative

Sample Output:

Recommended Fertilizer: Urea
Soil Health: Moderate
Climate Insight: Climate conditions are suitable.



AI Fertilizer Advisor

Predict the best fertilizer using Machine Learning, soil nutrients, crop stage, and real-time weather data. Built for smart farming decisions.

- ✓ Soil Analysis
- ✓ Weather Intelligence
- ✓ Crop Stage Optimization
- ✓ Cost Prediction

Enter Soil Details

City

Raipur

Soil pH

6.5

Nitrogen

45

Phosphorus

50

Potassium

60

Crop Type

Rice

Growth Stage

Vegetative

Predict Fertilizer



AI Fertilizer Prediction Report

Smart analysis using soil nutrients, crop growth stage, and live climate conditions

Recommended Fertilizer

Urea

Soil Health Status

Moderate

Estimated Cost

500

Selected City

Raipur

Explainable AI Analysis

Soil nutrients are balanced for Vegetative. Urea recommended.

Climate Insight

High temperature detected. Proper irrigation recommended.

Predict Again

View Weather Dashboard



Smart Climate Dashboard

Real-time agricultural weather intelligence powered by live weather API



Temperature

42.08°C

Current atmospheric temperature



Humidity

12%

Moisture level in air



Rainfall

0 mm

Recent precipitation data



AI Climate Advisory

High temperature detected. Proper irrigation recommended.

Raipur

New Prediction

← Back

Future Enhancements

- Add AI Chatbot for farmers
- Develop Android/Flutter mobile application
- Integrate IoT soil sensors

- Add multilingual support
- Use Deep Learning models
- Deploy system on cloud platform
- Add fertilizer recommendation ranking system

Gantt Chart

1. Planning & Structure

- Requirement analysis (including climate, crop stage, cost factors)
- Dataset collection (soil + weather + crop data)
- System design (ML model + rule-based + explainable AI modules)

2. Time Management

Week 1: Data Collection & Preprocessing

- Collect soil dataset (N, P, K, pH)
- Integrate weather data (temperature, humidity, rainfall)
- Data cleaning and feature engineering

Week 2: Model Building

- Train ML model (Decision Tree / Random Forest)
- Add climate-aware logic
- Validate model accuracy

Week 3: Advanced Logic Development

- Implement crop growth stage-based recommendation
- Develop cost optimization module
- Add explainable AI (reason generation)

Week 4: UI Development, Testing & Documentation

- Build user interface (Flask)
- Integrate all modules
- Test system with sample inputs
- Prepare documentation and report



Conclusion

The AI Fertilizer Recommendation System successfully demonstrates the integration of Machine Learning, Flask, and Weather API for smart agriculture solutions. The project provides an intelligent and user-friendly platform for fertilizer recommendation and soil health analysis. The system improves decision-making and supports precision farming.

References

- [1] Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media, 2019.
- [2] Flask Documentation, <https://flask.palletsprojects.com>

- [3] OpenWeatherMap API Documentation, <https://openweathermap.org/api>
- [4] Scikit-learn Documentation, <https://scikit-learn.org>
- [5] Kaggle, "**Fertilizer Recommendation Dataset**," Kaggle. [Online]. Available: <https://www.kaggle.com>
- [6] R. Singh, "Smart Agriculture using AI," in Proc. IEEE Conf. on Smart Systems, 2022, pp. 50-55.
- [7] A. Sharma and R. Gupta, "Machine Learning in Agriculture," International Journal of AI, vol. 5, no. 2, pp. 45-50, 2020.